1. Introduction

Aging infrastructure worldwide including bridges, industrial plants, tunnels, and civil constructions; requires frequent inspection and maintenance to prevent catastrophic failures. The manual execution of these tasks exposes human workers to hazardous conditions, incurs high operational costs, and is limited in coverage and repeatability *[Halder et al., 2023; Wang et al., 2025]*. Autonomous robotic inspection has therefore emerged as a critical technology, offering scalability, continuous operation, and the capacity to acquire high-precision data in environments that are inaccessible or dangerous for humans *[Halder et al., 2025]*. The integration of autonomous sensing platforms into non-destructive testing (NDT) workflows has been identified as a key enabler for next-generation infrastructure monitoring, combining visual detection, inertial navigation, and point-cloud mapping into a unified inspection pipeline.

Among existing robotic platforms, legged robots (and quadrupeds in particular) have proven the most effective for semi-structured environments that combine flat corridors with obstacles, stairs, and uneven ground. *ANYmal*, developed at ETH Zürich, was specifically designed for harsh industrial environments and has demonstrated reliable operation in oil and gas plants and search-and-rescue scenarios *[Hutter et al., 2017]*. More recently, Team CERBERUS deployed a multi-agent system of ANYmal robots augmented with LiDAR, cameras, and IMU; to win the DARPA Subterranean Challenge, establishing the quadruped as the reference platform for autonomous exploration in confined and semi-structured spaces *[Tranzatto et al., 2022]*. Despite these advances, quadrupedal robots face a fundamental efficiency limitation: walking gaits are energetically costly on flat surfaces, which restricts their operational autonomy for tasks that combine open traversal (e.g., factory floors) with the negotiation of obstacles or slopes *[Vogel et al., 2024]*. This trade-off between terrain adaptability and locomotion efficiency motivates the development of hybrid platforms capable of switching between legged and wheeled modes according to terrain properties.

Wheeled-legged (*wheeled-legged*) robots address this limitation by integrating actuated limbs (providing the dexterity required to negotiate irregular or sloped terrain) with wheels that constitute the most efficient effector on flat surfaces *[Bjelonic et al., 2022 survey]*. The ANYmal-on-wheels family, developed at ETH Zürich, is the most representative line in this category. Bjelonic et al. *[2019]* introduced the first whole-body controller capable of simultaneously combining walking and rolling with non-steerable torque-controlled wheels; subsequent work *[Bjelonic et al., 2020]* extended this to online trajectory optimization for hybrid maneuvers validated in the DARPA SubT, and Bjelonic et al. *[2022]* later incorporated offline motion libraries and MPC to achieve complex skills such as stair climbing and dynamic 180° turns. The most recent and impactful result in this family, Lee et al. *[2024, Science Robotics]*, demonstrated a complete pipeline of adaptive locomotion control, mobility-aware local planning, and city-scale global navigation, via deep reinforcement learning (RL) with privileged learning, achieving smooth walking-to-driving transitions as an emergent property of a single learned policy. The CENTAURO platform from IIT *[Kashiri et al., 2019; Klamt et al., 2019]*, a centaur-type quadruped with steerable omnidirectional wheels and a high-payload manipulator arm, offers a complementary design point with explicit disaster-response validation and provides the best published quantitative evidence for the energy/terrain trade-off that motivates a hybrid robot. At the compact end of the spectrum, Ascento *[Klemm et al., 2019, 2020]* is a bipedal wheeled-legged platform for indoor inspection, demonstrating that the design space spans from bipedal (Ascento) to centaur (CENTAURO) and quadrupedal (ANYmal-wheels) morphologies. Recent platforms such as Max *[Liu et al., 2024]*, FLORES *[Cui et al., 2025]*, and R-Taichi *[Jiang et al., 2025]* further expand the design space with alternative articulation schemes and RL-based controllers, confirming that the field is converging on hybrid locomotion as the dominant paradigm for versatile robot operation.

A critical observation emerges from this body of work: in all published wheeled-legged systems, locomotion mode switching is resolved either heuristically through hand-coded threshold logic *[Bjelonic et al., 2019; Kashiri et al., 2019],* by model predictive control *[Bjelonic et al., 2020; 2022]*, or by end-to-end RL policies that encode switching implicitly as a black-box behavior *[Lee et al., 2024; Humphreys & Zhou, 2025]*. None of these approaches employs a neurobiologically interpretable arbitration architecture. The basal ganglia, a subcortical neural circuit present in all vertebrates, are the canonical neural substrate for action selection *[Gurney et al., 2001a,b]*: through a salience-driven winner-take-all (WTA) mechanism implemented via disinhibition, they select and sustain a single behavioral channel from a set of competing alternatives while suppressing all others. To the best of the authors' knowledge, no prior work has applied a basal-ganglia-type architecture to arbitrate among rolling, walking, and omnidirectional locomotion modes in a hybrid quadruped. This constitutes the principal novelty of the present contribution.

Neural control architectures seek to emulate the information-processing structures found in biological systems in order to generate adaptive and robust motor responses *[Mead, 1990; Ijspeert, 2008]*. The computational model of basal ganglia (BG) proposed by Gurney, Prescott, and Redgrave (the GPR model *[Gurney et al., 2001a,b])* reformulates action selection as the resolution of competition among salience-coded channels through a feedforward off-center/on-surround network with selection and control pathways. Prescott et al. *[2006]* provided the first embodied validation of the GPR model, implementing it in a mobile robot performing a foraging task with five competing behaviors, demonstrating clean switching, appropriate behavioral persistence, and conflict-resolution dynamics analogous to those observed in animals. Girard et al. *[2003]* compared BG-based action selection against a classical WTA and showed that the BG's thalamocortical loops confer specific adaptive properties such as avoidance of dithering and energetic savings, that a simple WTA cannot reproduce; this distinction is directly relevant to the present work. More recent updates to the GPR paradigm include a biologically parsimonious three-pathway model (Go/NoGo/hyperdirect) *[Baston & Ursino, 2015]* and a neurorobotic study of simulated dopamine modulation that shows how tonic dopamine levels control the speed and frequency of mode switching *[Prescott et al., 2024]*, opening a pathway to meta-control of switching aggressiveness in future work. For the rhythmic generation of the quadrupedal walking gait, the present platform integrates central pattern generators (CPGs), neural circuits that produce coordinated rhythmic motor patterns from simple inputs *[Ijspeert, 2008]*; this combination of BG arbitration above and CPG execution below forms a hierarchical neural stack that is anatomically plausible and computationally modular *[Bellegarda & Ijspeert, 2022; Manoonpand et al., 2013]*.

Against this background, the present work proposes a hybrid quadrupedal robotic platform with four three-degree-of-freedom limbs, each incorporating an omnidirectional wheel at the distal segment (tibia), enabling transitions among three discrete locomotion modes: quadrupedal walking, differential rolling, and omnidirectional motion *(mobile* mode, with wheels as the primary effectors, and *articulated* mode, with the limbs providing the primary actuation). The platform integrates a sensory suite comprising an IMU, a LiDAR, and a camera for environmental perception. A multilayer perceptron (MLP) processes sensory data to detect terrain instability and compute per-channel salience inputs to the basal-ganglia network, which in turn executes mode selection via a WTA disinhibition mechanism. The complete system is validated in simulation (Gazebo) and on a physical prototype, demonstrating robust autonomous switching across flat surfaces, sloped terrain, and obstacle-laden environments. Relative to the current state of the art, the proposed architecture offers three differentiating properties: (i) *biological interpretability:* the switching logic is grounded in an explicit, neuroscientifically validated model rather than a learned black-box policy; (ii) *modularity:* perception, arbitration, and motor generation are decoupled, facilitating individual upgrade; and (iii) *autonomous energy-aware mode selection*, leveraging the efficiency advantage of wheeled locomotion on flat surfaces while preserving legged capability for complex terrain, directly addressing the operational limitation documented in deployed inspection systems *[Hutter et al., 2017; Tranzatto et al., 2022]*.

Response to Referee — Corrections to the Introduction

The following paragraph summarizes the revisions made to the Introduction section in response to the reviewers' feedback, primarily addressing **Reviewer 1, Comment 1** (the most critical comment regarding the introduction, highlighted in the authors' working copy): *"The Introduction section requires significant improvement. The authors have not adequately developed the state of the art in related research. As a result, reviewers lack sufficient horizontal comparisons and cannot evaluate the advancement of the proposed method."* Additional relevant feedback from Reviewer 2 (Comment 1, novelty not sufficiently demonstrated) and Reviewer 3 (Comment 6, no comparison with the state of the art or benchmarks) was also incorporated.

Specifically, the following changes were made:

(1) Expanded state-of-the-art coverage with horizontal comparisons. The original introduction cited only a small number of hybrid locomotion platforms without systematically comparing their approach to mode switching. The revised introduction now covers the full lineage of ANYmal-on-wheels *[Bjelonic et al., 2019; 2020; 2022; Lee et al., 2024]*, CENTAURO *[Kashiri et al., 2019; Klamt et al., 2019]*, Ascento *[Klemm et al., 2019; 2020]*, and recent competitive platforms (Max, FLORES, R-Taichi), supported by the survey *[Bjelonic et al., 2022 survey]*. This enables an explicit horizontal comparison across three axes: type of switching mechanism (heuristic vs. MPC vs. RL vs. basal ganglia), interpretability, and number of supported locomotion modes.

(2) Novelty clearly articulated and positioned relative to the literature. A dedicated paragraph now explicitly identifies the gap in the literature: no prior published work has applied a basal-ganglia-type architecture to arbitrate among rolling, walking, and omnidirectional modes in a hybrid quadruped. The contrast with the closest competing approach (the RL black-box switching of Lee et al. [2024]) is made explicit in terms of interpretability and modularity.

(3) Clarification of "mobile" and "articulated" modes (Reviewer 1, Comment 2). The first paragraph of the original introduction used the terms "mobile" and "articulated" without sufficient definition. The revised introduction now explicitly defines these terms at first use: *mobile mode* refers to the wheeled locomotion configuration in which the omnidirectional wheels at the distal segment serve as primary effectors, and *articulated mode* refers to the legged configuration in which the limb joints drive propulsion. These definitions are embedded in the contribution paragraph to ensure clarity from the outset.

(4) Motivation grounded in the infrastructure inspection use case. The revised introduction opens with a motivation paragraph explicitly justified by the infrastructure inspection application, citing empirical studies on quadruped-assisted construction inspection *[Halder et al., 2023]*, autonomous infrastructure monitoring *[Wang et al., 2025]*, and structural crack detection *[Halder et al., 2025]*. This directly addresses the concern raised in Reviewer 1, Comment 5 regarding Section 2.2: the inspection use case is now established and justified in the introduction rather than asserted without support in the methodology section.

(5) Bio-inspired control section expanded and better referenced. The basal-ganglia control rationale now cites the canonical GPR model *[Gurney et al., 2001a,b]*, its embodied robotic validation *[Prescott et al., 2006]*, the explicit comparison with classical WTA *[Girard et al., 2003]*, and recent updates to the paradigm *[Baston & Ursino, 2015; Prescott et al., 2024]*. CPG integration is justified with reference to the canonical review *[Ijspeert, 2008]* and recent CPG-RL work *[Bellegarda & Ijspeert, 2022]*. Together, these additions provide the theoretical grounding requested by the reviewers and allow the reader to evaluate the proposed architecture against established alternatives.

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